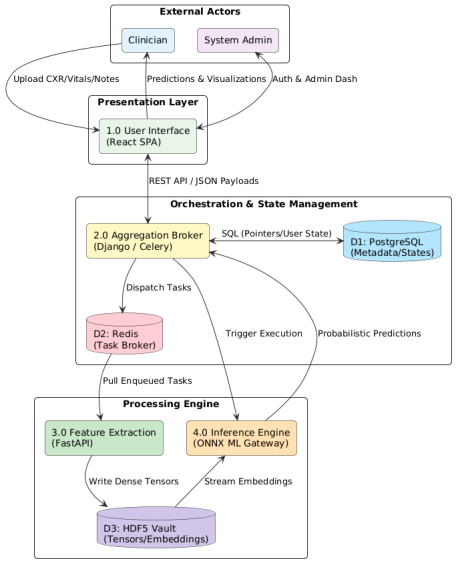
Causal Multimodal Medical Modelling System

Department of Computer Science and Engineering, FISAT

Introduction

- **Background**: ICUs generate vast multimodal data (vitals, labs, imaging, notes).
- **Importance**: Holistic synthesis is vital for early patient deterioration detection.
- **Challenges**: Existing models struggle to fuse temporal and spatial data inherently missing in clinical records.
- **Objective**: Engineer a causally-aware deep learning framework to aggregate multimodal data for severity clinical inferencing.

System Architecture

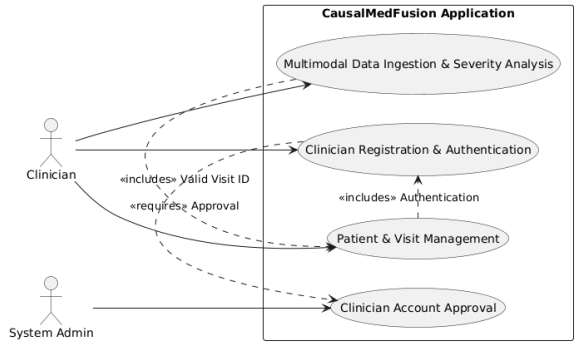


- **Decoupled Extraction**: FastAPIs independently process physiological measurements, text, and images.
- **Message Brokering**: Redis orchestrates asynchronous Celery tasks without UI blocking.

Proposed System

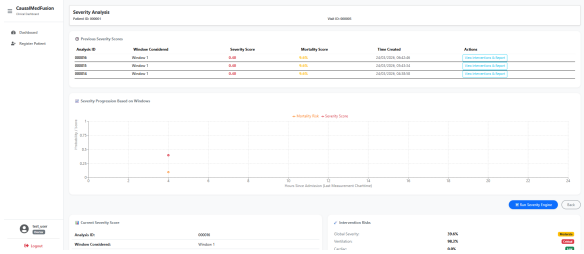
- **Overview**: Django-React framework integrating Transformer and CNN subgraphs.
- **Key Features**: Time-window batching and causal-masked attention modeling.
- **Advantages**: Resilient scaling and graceful operation against missing data.
- **Stack**: Python, React.js, PostgreSQL, Redis, Celery, ONNX Runtime.

Use Case Diagram



- **Actors Involved**: Clinicians, System Administrators.
- **Core Interactions**: Authenticating, managing records, uploading artifacts, viewing severity trajectories.

Results



- **Interactive Trajectories**: Dynamic rendering of unified severity scores across time-windows.
- **Intervention Risks**: Normalized probabilities predicting Mortality and Ventilation needs.
- **System Resilience**: High-throughput asynchronous batching operates without failing under stress.

Conclusion

- Successfully unified unstructured notes with formatted physiological arrays.
- Scalable microservices incorporate an ONNX Gateway resilient to data sparsity.
- Delivered an interactive UI providing actionable prognostics for rapid patient triage.

Future Scope

- **Expanded Signals**: Support continuous waveform inputs (ECG/telemetry streams).
- **Explainable AI**: Implement explicit SHAP attention-visualization over Chest X-Rays.
- **Federated Integration**: Broaden deployment to multi-region hospital databases.

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